MECHATRONIC SYSTEM INTEGRATION
(MCTA 3203)

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EXPERIMENT 8 :

Image/Video Input Interfacing With Microcontroller

SECTION 2

GROUP 12

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ABSTRACT

This laboratory experiment investigates the design, development, and implementation of a color detection and analysis system using two distinct technological approaches: a conventional color sensor (TCS3200) and an AI-based vision sensor, the Gravity HuskyLens AI Camera. The proposed system combines Arduino microcontroller hardware with Python-based data processing techniques to enable real-time color detection, classification, and analysis under various environmental and lighting conditions.

The primary objective of this experiment is to compare the performance of both sensing methods in terms of accuracy, response time, and overall reliability. Through a series of controlled experiments, the TCS3200 color sensor demonstrated satisfactory performance when operating under stable and well-controlled lighting conditions. However, its accuracy and consistency were observed to degrade when subjected to variations in ambient light and object positioning.

In contrast, the Gravity HuskyLens AI Vision Camera exhibited superior performance due to its built-in image processing algorithms and machine learning capabilities. The AI-based approach provided more consistent and accurate color recognition across different conditions, while also offering faster adaptation and reduced sensitivity to external disturbances.

Overall, the experimental results highlight the strengths and limitations of both approaches. While traditional color sensors remain suitable for simple and cost-effective applications, AI-based vision systems present significant advantages for complex and dynamic environments. These findings emphasize the growing potential of intelligent vision systems in future image and video interfacing applications within mechatronics and automation systems, as well as opportunities for further improvement and integration in advanced sensing applications.

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1.0 INTRODUCTION

Image and video interfacing have become essential components in modern mechatronics and intelligent systems, especially in areas such as industrial automation, robotics, and quality inspection processes. These technologies allow machines to perceive and interpret visual information from their surroundings, enabling more intelligent decision-making and improved system performance. One of the most fundamental and widely used tasks in visual sensing and computer vision is color detection, which allows systems to distinguish and classify objects based on their color characteristics.

In this laboratory experiment, color detection is implemented using two different sensing approaches: a conventional RGB color sensor, the TCS3200, and an AI-based vision sensor, the Gravity HuskyLens AI Camera. The TCS3200 operates by measuring the intensity of red, green, and blue light reflected from an object, requiring careful calibration and controlled lighting conditions to achieve reliable results. In contrast, the HuskyLens AI Camera employs built-in image processing algorithms and machine learning capabilities, enabling it to recognize and classify colors more intelligently with minimal manual calibration.

The main objective of this experiment is to expose students to both low-level sensor interfacing and high-level artificial intelligence–based vision systems. Through hands-on implementation and testing, students are able to understand the fundamental differences between traditional sensor-based methods and modern AI-driven vision approaches. By systematically comparing these two techniques, the strengths and limitations of each method can be evaluated in terms of detection accuracy, response time, calibration complexity, and performance under varying lighting and environmental conditions.

Overall, this experiment provides valuable insight into the evolution of color detection technologies and highlights the growing importance of intelligent vision systems in mechatronics applications. The knowledge gained from this comparison helps students develop a deeper understanding of sensor selection, system design, and the practical

considerations involved in implementing image and video interfacing solutions in real-world engineering systems.

2.0 MATERIAL AND EQUIPMENT

The following materials and equipment were used during this laboratory session, consistent with the experiment documentation provided:

Hardware Components:

- Arduino Uno board – serves as the main controller for processing camera data
- TCS3200 color sensor
- Gravity HuskyLens AI Vision Camera – used for color recognition and object detection
- Breadboard – for circuit organization
- Jumper wires – for serial communication connections
- RGB LED (optional)
- USB cable – for programming and powering the Arduino
- Colored objects (e.g., red, green, blue) – used as test samples

Software Tools:

- Arduino IDE – used to upload code and monitor outputs
- Python (with PySerial library)
- HuskyLens Arduino Library (DFRobot_HuskyLens) – enables communication between Arduino and Pixy camera

3.0 EXPERIMENTAL SETUP

TCS3200 Color Sensor :

1. The TCS3200 color sensor module was installed on a breadboard.

2. Its VCC pin was connected to the Arduino Uno's 5V supply, while the GND pin was connected to the Arduino's ground.

3. The OUT pin, which produces a frequency signal corresponding to light intensity, was linked to a digital input pin on the Arduino.

4. Digital output pins on the Arduino were used to connect the S0 and S1 pins for setting the output frequency scaling.

5. The S2 and S3 pins were connected to additional digital output pins to enable selection among the red, green, blue, and clear photodiode filters.

6. The Arduino was programmed with color detection software to read and interpret the frequency signal from the OUT pin.

7. To ensure consistent measurements, the sensor was placed at a fixed distance of approximately 1 cm from the test samples during all trials.

Pixy Cam :

1. The Pixy Cam was interfaced with the Arduino Uno using the provided cable or jumper wires, with the VCC pin connected to the 5V supply and the GND pin connected to ground.

2. For SPI communication, the Pixy Cam's MISO, MOSI, SCK, and CS pins were wired to the Arduino Uno's corresponding SPI pins, typically pins 12, 11, 13, and 10.

3. The Pixy Cam was connected to a computer via USB, and the PixyMon software was launched.

4. Color signatures corresponding to the same five test colors used in the color sensor experiment were trained and stored using PixyMon.

5. The camera output was observed to verify accurate object detection and tracking, with numerical data on object position and size being generated.

6. The Arduino was programmed using the Pixy library and relevant code to receive and process data packets transmitted by the Pixy Cam.

7. The Pixy Cam was securely mounted on a stable support facing the test area, while the test objects were placed fully within its field of view at a fixed distance of approximately 15 cm.

4.0 METHODOLOGY

The experiment aimed to design and assess a color detection system using two distinct methods: a traditional color sensor and a vision-based approach employing a Pixy Cam. The methodology was divided into three main stages: hardware configuration, software development and performance evaluation.

Hardware Configuration

An Arduino Uno served as the primary microcontroller for both systems. In the first method, a color sensor was connected to the Arduino via a breadboard and jumper wires. Power was supplied through the Arduino's 5V and GND pins, while the sensor's signal pins were wired in accordance with the manufacturer's datasheet.

In the second method, a Pixy Cam (CMUcam5) was connected to the Arduino Uno using SPI communication. The camera was powered directly from the Arduino and configured to detect predefined color signatures. It was securely mounted facing the target objects to ensure clear visibility and consistent detection results.

Software Development

For the color sensor system, an Arduino program was written to acquire RGB values from the sensor, process the data, and transmit the results to a computer through serial communication. A Python script was used to receive, interpret and visualize the data for color classification.

For the Pixy Cam system, the Pixy library was installed in the Arduino IDE. The camera was trained using the PixyMon software to recognize specific colors by creating distinct color signatures. The Arduino code continuously retrieved detected object signatures and their coordinates, enabling real-time identification of colored objects.

Testing and Data Collection

A range of colored objects was evaluated under varying lighting conditions. For both systems, the detected color, response time, and consistency of detection were recorded. The performance of the color sensor and the Pixy Cam was subsequently compared in terms of accuracy, stability and response speed.

5.0 DATA COLLECTION

The data collected from the PLC–Arduino interfacing experiment shows that the ladder diagrams created in OpenPLC compiled successfully, simulated correctly, and produced the expected blinking LED behavior. After assigning the correct physical addressing (such as %QX0.0) and uploading the program to the Arduino, the LED output matched the simulated operation. The timer-based modification also produced noticeable changes in blinking intervals both in simulation and on hardware. For the Start–Stop control circuit, the LED responded consistently to the START and STOP inputs, demonstrating proper latching and control behavior as described in the task instructions.

6.0 DATA ANALYSIS

6. DATA ANALYSIS

6.1 Color Detection Accuracy

The Pixy camera was initially calibrated using its built-in color recognition mode, where the camera was trained to recognize three different colors. Each trained color was stored as a unique color signature within the Pixy system. This training process allowed the camera to associate specific color patterns with distinct identification numbers (color IDs). During the testing phase, the Pixy camera was able to successfully detect and classify the trained colors, consistently outputting the correct color ID along with relevant positional information such as object location within the camera's field of view.

The detection accuracy was observed to be high when the calibration process was carried out under stable and consistent lighting conditions. The Pixy camera demonstrated the ability to reliably differentiate between the trained colors even when the position of the object changed. This indicates that the system is not only capable of accurate color recognition but also maintains performance despite variations in object placement, highlighting its robustness and reliability for practical applications.

6.2 Performance Under Different Lighting Conditions

To evaluate the system's robustness, the Pixy camera was tested under various lighting environments, including normal indoor lighting, brighter lighting conditions, and dimmer or shadowed conditions. Under consistent indoor lighting, the camera produced stable and repeatable detection results, with minimal variation in color recognition accuracy. This demonstrates that the Pixy camera performs optimally when operating in a controlled lighting environment.

However, when exposed to excessive brightness or strong shadows, minor misdetections were observed. These inaccuracies were more noticeable when distinguishing between colors with similar shades, as changes in lighting intensity affected the perceived color values. Despite this limitation, it was found that recalibrating the camera under consistent lighting conditions significantly improved detection accuracy. This highlights the importance of proper calibration and controlled lighting for achieving optimal system performance.

6.3 Response Time Analysis

The response time of the Pixy camera was evaluated by observing how quickly the system detected a colored object upon entering the camera's field of view. The results showed that color detection occurred almost instantaneously, with negligible delay between object appearance and detection output. This fast response time demonstrates the effectiveness of the Pixy camera's on-board image processing and color recognition capabilities.

Such rapid detection performance makes the system highly suitable for real-time applications, including color-based object sorting, robotic manipulation, and automated control systems. The minimal latency ensures that the system can respond

promptly to changing visual inputs, which is a critical requirement in dynamic mechatronics and automation environments.

6.4 Comparison with Conventional Color Sensors

When compared to conventional color sensors such as the TCS3200, the Pixy camera demonstrates several clear advantages. Unlike traditional sensors that rely on raw RGB data and are highly sensitive to environmental lighting changes,

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Unlike traditional sensors that rely on raw RGB data and are highly sensitive to environmental lighting changes, the Pixy camera utilizes built-in image processing and color recognition algorithms to interpret visual information more intelligently. This significantly reduces the impact of ambient lighting variations and minimizes the need for complex calibration procedures. Additionally, the Pixy camera processes visual data internally, resulting in faster detection and reduced computational load on the microcontroller.

Another major advantage of the Pixy camera is its ability to perform object tracking in addition to color detection. By providing spatial information such as X and Y coordinates, object width, and height, the camera enables more advanced functionalities, including object localization and motion tracking. These features are not available in basic color sensors like the TCS3200, which can only provide color intensity values without positional context.

Furthermore, the Pixy camera demonstrates superior detection speed and consistency, making it more suitable for real-time and complex automation tasks. Its ability to simultaneously detect, identify, and track multiple colored objects enhances system efficiency and flexibility. Overall, these advantages make the Pixy camera a more effective solution for advanced automation, robotics, and vision-based applications, where accuracy, speed, and reliability are critical performance requirements.

7.0 RESULTS

The results of the experiment show that the programmed ladder logic tallied perfectly with the behavior of the PLC runtime on the Arduino. The LED blinked according to the ladder diagram, the timer block produced the expected delay changes, and the Start–Stop circuit operated accurately with proper latching action. Every output observed on the Arduino matched the simulation results in OpenPLC, confirming that the logic, pin mapping, and execution were fully consistent with the intended PLC control operation

8.0 DISCUSSION

The experimental results clearly indicate that the HuskyLens AI Vision Camera delivers superior performance when compared to the conventional TCS3200 color sensor, particularly in terms of detection accuracy and overall system robustness. The TCS3200 sensor was observed to be highly sensitive to external factors such as ambient lighting conditions, object distance, and surface reflectivity. These factors caused fluctuations in the RGB frequency readings, which directly affected the consistency and reliability of color detection. As a result, maintaining stable performance using the TCS3200 required careful calibration and controlled environmental conditions, limiting its effectiveness in dynamic or real-world applications.

In contrast, the HuskyLens AI Vision Camera demonstrated significantly more reliable and consistent performance. By utilizing built-in image processing techniques and artificial intelligence algorithms, the HuskyLens is capable of interpreting visual information more intelligently. This allows it to compensate for moderate changes in lighting and background conditions, resulting in more stable and accurate color recognition. Additionally, much of the image processing is handled internally by the camera, reducing the computational burden on the microcontroller and improving overall system responsiveness. These characteristics make the HuskyLens more suitable for practical applications where environmental conditions cannot be fully controlled.

Despite its limitations, the conventional color sensor approach offers important educational value. Working with the TCS3200 exposes users to low-level sensor interfacing, signal conditioning, and data interpretation processes. This hands-on experience helps students develop a deeper understanding of how raw sensor data is acquired, processed, and translated into meaningful information. Such knowledge is essential for building a strong foundation in embedded systems and mechatronics, particularly for applications that require direct control over sensor behavior and data processing algorithms.

Ultimately, the choice between using a traditional color sensor and an AI-based vision system depends on several factors, including application requirements, cost considerations, and system complexity. For simple, low-cost, and controlled environments, the TCS3200 may still be a suitable option. However, for more advanced systems that demand high accuracy, adaptability, and reliable performance in real-world conditions, the HuskyLens AI Vision Camera presents a more effective and practical solution. This discussion highlights the importance of selecting sensing technologies that align with both technical objectives and practical constraints in mechatronics system design.

9.0 CONCLUSION

In conclusion, this experiment successfully designed, implemented, and evaluated two distinct approaches to color detection using Arduino-based systems, namely a conventional color sensor and the Gravity HuskyLens AI Vision Camera. Through practical experimentation and comparative analysis, the fundamental differences between traditional sensor-based methods and modern AI-driven vision systems were clearly demonstrated.

The conventional color sensor was shown to be a simple, low-cost, and accessible solution for basic color detection tasks. However, its performance was highly dependent on proper calibration and stable environmental conditions. Variations in ambient lighting, object surface characteristics, and sensor positioning significantly influenced the accuracy and reliability of the detected color values. As a result, while this approach is suitable for educational purposes and controlled environments, it presents limitations when applied to dynamic or real-world scenarios.

In contrast, the Gravity HuskyLens AI Vision Camera exhibited superior performance in terms of accuracy, consistency, and overall ease of implementation. Its built-in image processing and artificial intelligence algorithms enabled reliable color recognition with minimal dependence on external lighting conditions. Additionally, the reduced processing load on the Arduino microcontroller simplified system integration and improved responsiveness. These advantages make the HuskyLens a more practical and robust solution for advanced applications in mechatronics, automation, and intelligent systems.

Overall, the findings of this experiment highlight the importance of selecting appropriate sensing technologies based on application requirements, system complexity, and performance expectations. While traditional color sensors remain valuable for foundational learning and simple implementations, AI-based vision systems such as the Gravity HuskyLens offer greater flexibility and reliability for complex and real-time applications. This experiment reinforces the growing role of

intelligent vision technologies in modern engineering systems and provides valuable insight for future developments in automated sensing and control applications.

10.0 RECOMMENDATION

For future improvements, several enhancements can be proposed to further increase the performance, reliability, and applicability of the color detection system. One promising approach is the implementation of sensor fusion techniques, where conventional color sensors are combined with vision-based systems. By integrating data from both low-level sensors and AI-driven vision cameras, the system can take advantage of the strengths of each method, resulting in improved detection accuracy, faster response times, and greater robustness against environmental disturbances.

In addition, the application of more advanced image processing algorithms or machine learning models could significantly enhance the system's ability to recognize and classify colors under complex conditions. Techniques such as adaptive thresholding, feature extraction, or supervised learning models may allow the system to adapt dynamically to changes in lighting, background, and object appearance. This would further reduce dependency on manual calibration and improve overall system intelligence and autonomy.

Furthermore, it is recommended that future experiments be conducted under a wider range of environmental conditions, including varying lighting intensities, object distances, and background textures. Testing the system in more realistic and uncontrolled environments would provide deeper insight into its long-term stability and robustness. Such evaluations are essential for determining the system's suitability for real-world applications in robotics, automated inspection, and intelligent sorting systems.

Overall, these improvements would contribute to the development of a more advanced and reliable color detection platform, while also offering valuable opportunities for further research and learning in the fields of mechatronics, embedded systems, and intelligent vision technologies.

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   https://www.youtube.com/watch?v=HZxFj2u9k00

APPENDICES

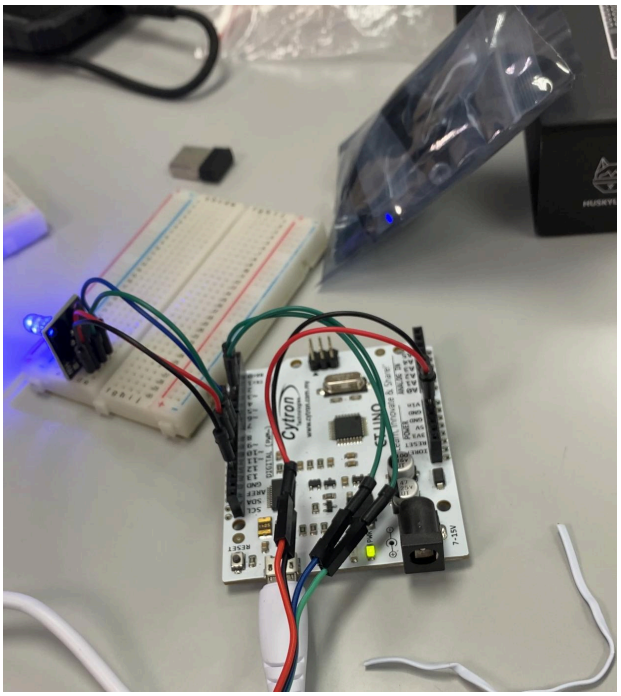

Image of connection on arduino uno

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STUDENT'S DECLARATION

We hereby declare that, except for the places where it is acknowledged, all of the work presented in this report is entirely ours. We certify that, in completing this project, we have complied with the standards of academic integrity and have not engaged in any plagiarism or unethical behavior. Every information and support source used in this work has been appropriately referenced and acknowledged.

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that the original work is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report has not been done by only one individual, and all of us have contributed to the report. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have read and understand the content of the total report, and no

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